(Square)

1

HW

The difference of the areas of two squares drawn on two line segments of different lengths is $32\ cm^2$. Find the length of the greater line segment if one is longer than the other by 2 cm

a) 7 b) 9

c) 11 d) 16

2

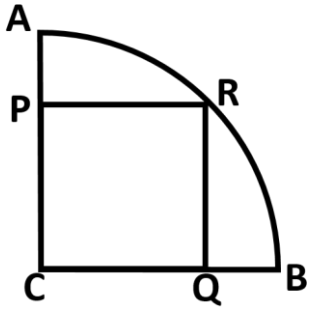
HW

The perimeters of 5 squares are $24\ cm, 32\ cm, 40\ cm, 76\ cm$ and 80 cm respectively. The perimeter of another square equal in area to sum of the area of these squares is:

5

a) 31 **b) 62**

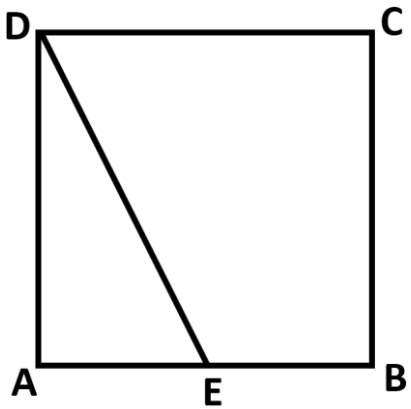
c) 124 **d) 961**



3) In the adjoining figure a quadrant (of circle) inscribes a square of maximum possible area. If the radius of the circle be r then what is the area of the square?

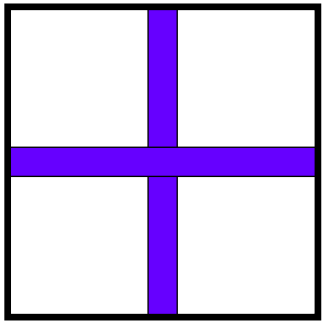
a) $\frac{r}{2}$ b) $\frac{3r}{5}$

c) $\frac{r^2}{\sqrt{3}}$ d) $2\sqrt{6}r$



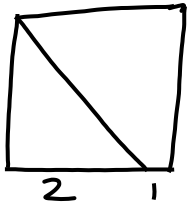
(4) ABCD is a square, E is a point on AB such that $BE = 17cm$. The area of triangle ADE is 84 cm^2. What is the area of square (in cm^2)?

a) 400 b) 625
c) 729 d) 576



5 A square garden has two roads passing through its middle, both parallel to its sides. Each side of square is 40 m in length and the width of roads is 10 m. Find the cost of graveling the roads at the rate of Rs.25 per m^2.

a) 17500 b) 7000
c) 700 d) 20000



6 **If the ratio of areas of two squares is $225:256$, then the ratio of their perimeter is**

a) 225:256 b)256:225
c) 15:16 d) 16:15

7 The diagonal of a square is $4\sqrt{2}$ cm. The diagonal of another square whose area is double that of the 1st square is

a) $8\sqrt{2}$ b) 16

c) $\sqrt{32}$ d) 8

8 **The diagonal of a square A is $(a+b)$. The diagonal of a square whose area is eight times the area of square A,**

a) $4(a+b)$ b) $8\,(a+b)$

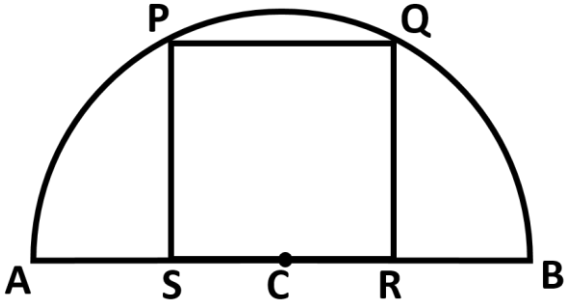
c) $2\sqrt{2}(a+b)$ d) $2\sqrt{2}(a-b)$

(9) The diagonal of a square A is $(a + b)$ units. What is the area (in square units) of the square drawn on the diagonal of square B whose area is twice the area of A?

a) $4(a + b)$ b) $(a + b)$

c) $8(a + b)^2$ d) $2(a + b)^2$

CGL 2019 T1



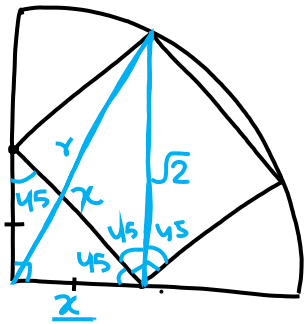
10 **In the adjoining figure PQRS is a square of maximum possible area which is circumscribed by the semicircle. Points R and S lie on the diameter AB. What is the area of the square if the radius of the circle is r?**

a) $\frac{\sqrt{3}}{4}\ r^2$

b) $\frac{4}{5}\ r^2$

c) $\frac{3}{5}\ r^2$

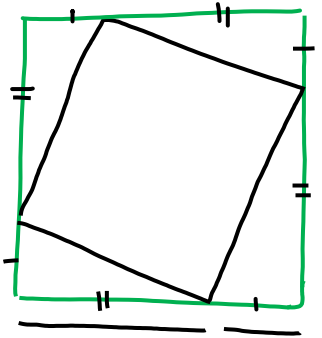
d) $\frac{\sqrt{5}}{4}\ r^2$

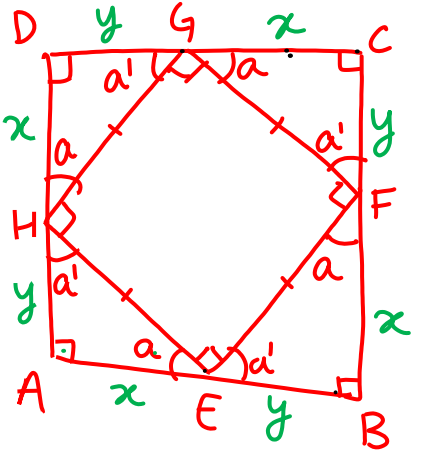


11 **A square is inscribed in quarter-circle in such a manner that two of its adjacent vertices lie on the two radii at an equal distance from the centre, while the other two vertices lie on the circular arc. If the square has sides of length x, then the radius of the circle is**

a) $\frac{2x}{\pi}$ b) $\frac{\sqrt{5}x}{\sqrt{2}}$

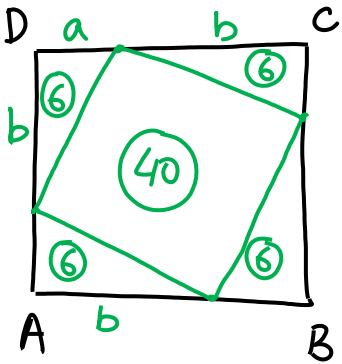
c) $\frac{16x}{\pi+4}$ d) $\sqrt{2}x$



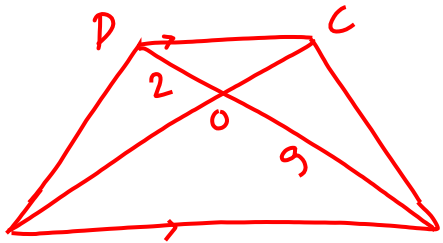

D
y
G
x
C
a'
a
x
a
a'
y
H
F
y
a'
a
x
a
a'
A
x
E
y
B

(12) Points E, F, G and H lie on the sides AB, BC, CD, and DA, respectively, of a square $ABCD$. If $EFGH$ is also a square whose area is 62.5% of that of $ABCD$ and CG is longer that EB, then the ratio of length of EB to that of CG is

a) $2:5$ b) $4:9$
c) $3:8$ d) $1:3$



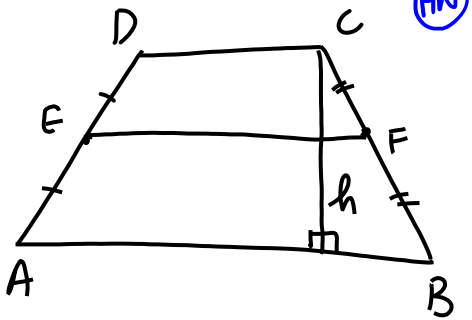
Trapezium



13 HW **In a trapezium, one diagonal divides the other in the ratio 2:9, If the length of the larger of the two parallel sides is 45cm, then what is the length (in cm) of the other parallel sides?**

a) 10 **b) 5**

c) 18 **d) 14**

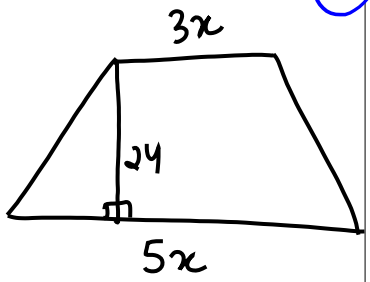


M HW **The length of the line formed by joining the mid points of non-parallel sides of a trapezium is 40cm and the shortest distance between parallel sides is 12 cm. Find the area of trapezium.**

a) 480 **b) 240**
c) 400 **d) 960**

15 HW

The area of a field in the shape of trapezium measures $1440\ m^2$. The perpendicular distance between its parallel sides is 24 m. If the ratio of parallel sides is 5:3, the length of longer side is

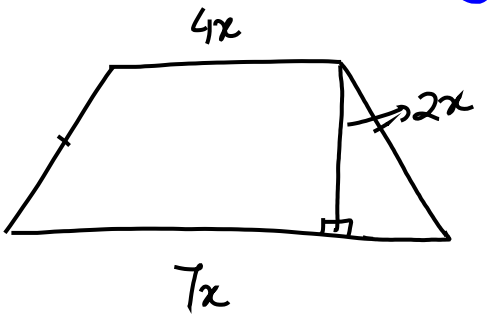


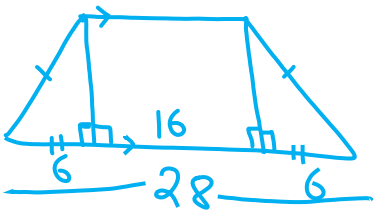
a) 15 b) 75
c) 45 d) none

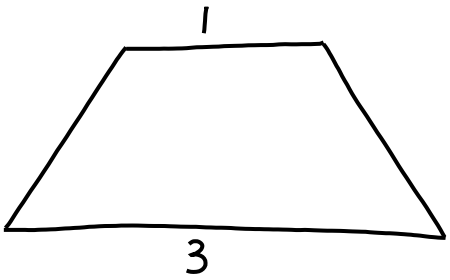
(16) The area of an isosceles trapezium is 176 cm^2 and the height is $\frac{2}{11}th$ of the sum of its parallel sides. If the ratio of the length of the parallel sides is 4 : 7, then the length of a diagonal (in cm) is

a) $4\sqrt{137}$ b) $\sqrt{137}$

c) $2\sqrt{137}$ d) 24





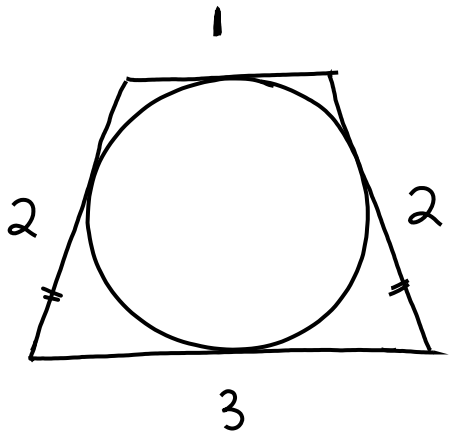


(17) **The length of parallel sides of a trapezium is in ratio 1:3 while non-parallel sides ratio is 2:3 if the ratio of length of larger parallel side to the larger non-parallel side is 2:1 and height is $\frac{15\sqrt{15}}{4}$. Find the area of the trapezium.**

a) $120\sqrt{15}$ **b)** $130\sqrt{15}$

c) $150\sqrt{15}$ **d)** $160\sqrt{15}$

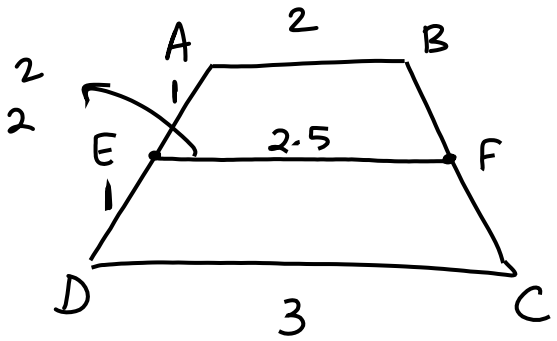
$\sqrt{}$



(12) An isosceles trapezium circumscribes a circle. One of the parallel sides of the trapezium is three times as long as other and its perimeter is $40cm$. Then area of the trapezium is ____ sq.cm.

a) 50 b) $50\sqrt{3}$

c) $50\sqrt{2}$ d) 60

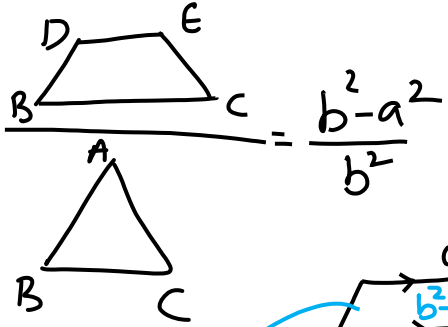


(19) $ABCD$ is a trapezium with parallel sides $AB = 2cm$ and $DC = 3\ cm$. E and F are the mid-points of the non-parallel sides. The ratio of area of $ABFE$ to area of $EFCD$ is:

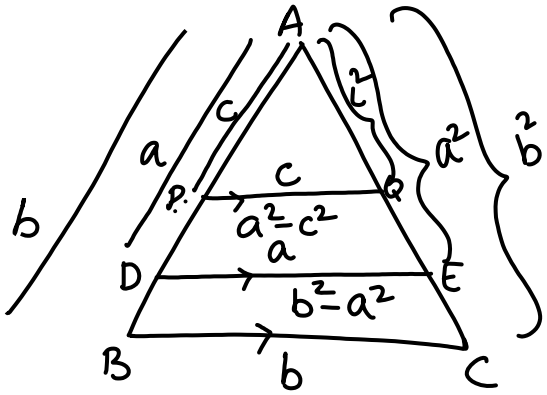
a) $9:10$ b) $8:9$

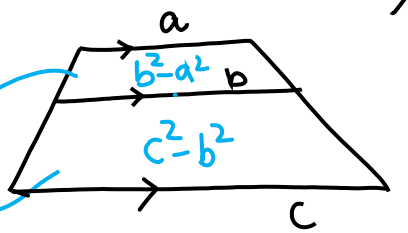
c) $9:11$ d) $11:9$

CDS 2013 I

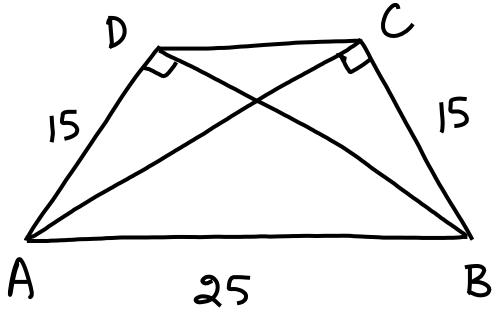


$$\frac{\text{BC}}{\text{DE}} = \frac{b^2-a^2}{b^2}$$



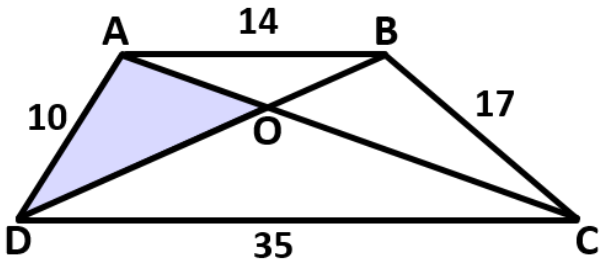


Ratio $\frac{b^2-a^2}{c^2-b^2}$



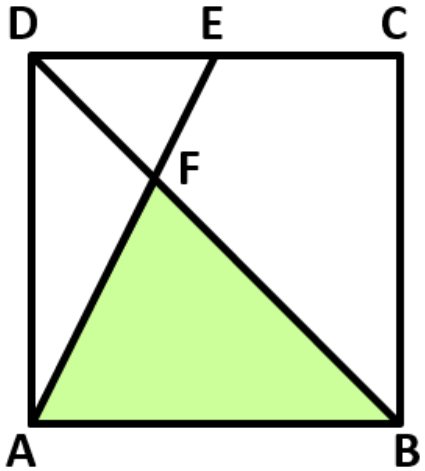
20 **In a trapezium ABCD, AB is parallel to CD. BD is perpendicular to AD. AC is perpendicular to BC. If AD=15 cm and AB=25cm, then the area of the trapezium is:**

a) 192cm² **b) 172cm²**
c) 232cm² **d) None of these**



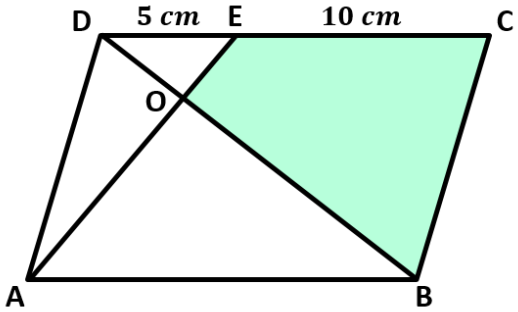
21 In the given figure, the diagonals of trapezium ABCD intersect at O. Find the area of the shaded region according to the lengths of the sides mentioned in the figure.

a) 20 b) 30
c) 40 d) 10



22 In the given figure, ABCD is a square and E is the mid point of DC. What is the ratio of areas of shaded and unshaded portion?

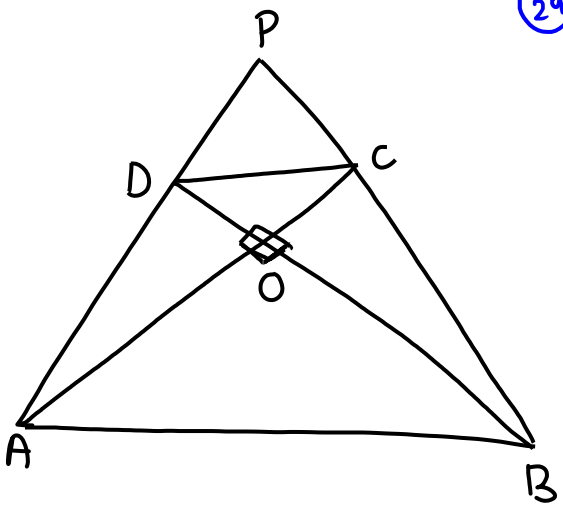
a) $1:3$ b) $2:3$
c) $1:2$ d) $3:4$



(23) In the given figure, what fraction of the area of the parallelogram ABCD is shaded ?

a) $\frac{11}{24}$ b) $\frac{1}{3}$

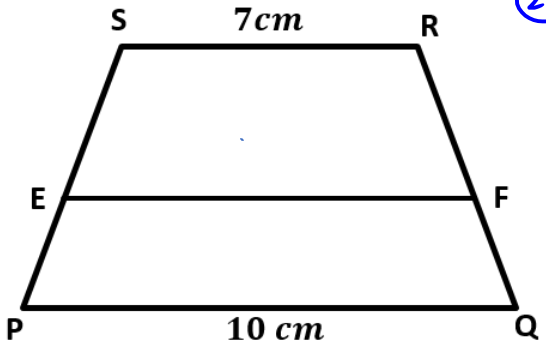
c) $\frac{11}{18}$ d) $\frac{2}{5}$



(24) **The diagonals of a trapezium are at right angles and the slant sides, if produced, form an equilateral triangle with the greater of the two parallel sides. If the area of the trapezium is 16 sq cm, then the distance between the parallel side is**

a) 2 cm **b) 4 cm**
c) 8 cm **d) CND**

CDS 2015 I



25 In the given figure, the area of trapezium $PQFE$ is half of the area of trapezium $EFRS$. SR and PQ is $7cm$ and $10cm$ respectively, then find the length of EF.

a) $9cm$ b) $2\sqrt{21}$cm

c) $4\sqrt{6}$cm d) $\sqrt{83}$cm